

## SEMESTER 1 EXAMINATIONS 2021/2022

**MODULE:** CA119 - Data Science and Databases

**PROGRAMME(S):**

|       |                                        |
|-------|----------------------------------------|
| DS    | BSc in Data Science                    |
| AP    | BSc in Applied Physics                 |
| ECSAO | Study Abroad (Engineering & Computing) |

**YEAR OF STUDY:** 1,3,O

**EXAMINER(S):**

|                             |            |            |
|-----------------------------|------------|------------|
| Prof. Mark Roantree         | (Internal) | (Ext:5636) |
| Assoc. Prof. Sheila McBreen | (External) | External   |
| Prof. Gerard O'Connor       | (External) | External   |

**TIME ALLOWED:** 2 Hours

**INSTRUCTIONS:** Answer all 4 questions. All questions carry equal marks.

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**PLEASE DO NOT TURN OVER THIS PAGE UNTIL YOU ARE INSTRUCTED TO DO SO.**

The use of programmable or text storing calculators is expressly forbidden.

Please note that where a candidate answers more than the required number of questions, the examiner will mark all questions attempted and then select the highest scoring ones.

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*There are no additional requirements for this paper.*

**Very Important First Step: Select your Company.**

| <b>ExamCode</b> | <b>Company</b>              |
|-----------------|-----------------------------|
| 0416            | Hertz Car Rentals           |
| 0931            | Ryanair                     |
| 1354            | Aer Lingus                  |
| 1492            | Local Tennis Club           |
| 1562            | Elverys Sports              |
| 1706            | DCU Library                 |
| 1752            | Allied Irish Banks          |
| 1884            | Packers NFL Football Team   |
| 1931            | Delta Airlines              |
| 2192            | Petworld                    |
| 2274            | Woodies DIY Store           |
| 2343            | Lifestyle Sports            |
| 2396            | Bord Failte                 |
| 2486            | Maxol Garages               |
| 2793            | Irish Rail                  |
| 3061            | St. James Hospital          |
| 4256            | DCU                         |
| 4881            | Mercedes Benz               |
| 5523            | Dawn Meats                  |
| 5632            | Local Soccer Club           |
| 6054            | ODEON Cinema                |
| 6173            | Diageo                      |
| 6323            | Irish Ferries               |
| 6876            | Gaelic Athletic Association |
| 7073            | Irish Boxing Association    |
| 7166            | Bus Eireann                 |
| 7181            | Bank of Ireland             |
| 7296            | Petworld                    |
| 7453            | VIP Taxi Company            |
| 7739            | Dublin Airport              |
| 9186            | Milanos Restaurant          |
| 9421            | Paint Suppliers             |
| 9553            | Eason's Bookstore           |

**On your Answer Book, Write: your name; Student ID, ExamCode, Company.**

**QUESTION 1****[TOTAL MARKS: 25]****Entity Relationship Modelling**

Design an Entity-Relationship model for your company, but do it in steps according to parts, a, b and c below. You will be required to provide 4 entities. Ensure that they are important entities for your company.

**Q 1(a)****[8 Marks]**

Provide 4 entities, in *separate* Entity-Relationship diagrams for each entity. Ensure that you include at least 1 derived attribute; 2 multi-valued attributes; 2 composite attributes; and 1 complex attribute.

Underneath each E-R diagram, write the names and types of the 4 different attributes. Ensure that they are properly represented in the E-R diagram.

**Q 1(b)****[9 Marks]**

Provide one example of each of the following relationships: one-to-one, one-to-many, and many-to-many. Draw a new E-R diagram to represent these relationships together with the entities from part a)

.

**Q 1(c)****[8 Marks]**

In binary relationships, there are 2 types of constraint: cardinality ratio and participation.

Provide an example of each constraint type from your relationship examples in part (b) of this question. Explain why your two chosen examples require these constraints.

**QUESTION 2**  
*SQL Programming*

**[TOTAL MARKS: 25]**

This question assumes that your E-R diagram is then converted into a set of tables. This question requires you to provide 5 SQL queries for your company database.

You will be awarded marks for the usefulness of your query expression. Parts *a* and *b* allow for relatively simple queries. Parts *c,d* and *e* require more complex queries and will be marked accordingly.

**For ALL questions, you must write the query in English first and then write the SQL version.**

**Q 2(a)** **[3 Marks]**

Provide a SQL expression that uses a projection (a subset of attributes) and selection (a filter condition) for one of your entities.

**Q 2(b)** **[4 Marks]**

Provide a SQL expression that computes a dynamic attribute from one of your entities. In your query, also use an aggregation such as SUM or AVERAGE.

**Q 2(c)** **[4 Marks]**

Provide a SQL expression that JOINS 3 tables.

**Q 2(d)** **[6 Marks]**

Provide a SQL expression that JOINS 2 entities and also uses a GROUP-BY statement.

**Q 2(e)** **[8 Marks]**

Provide a SQL expression that takes either 2c or 2d and instead uses a nested query.

### QUESTION 3

[TOTAL MARKS: 25]

#### *Functional Dependencies*

Functional dependencies take the form  $A \rightarrow B$ , where  $A$  and  $B$  have one or more attributes.

#### **Q 3(a)**

**[3 Marks]**

Provide a sample table for one of your entities (from questions 1 and 2) and enter 10 rows of data.

#### **Q 3(b)**

**[6 Marks]**

Using the sample table, provide 2 functional dependencies. Explain what these dependencies mean.

#### **Q 3(c)**

**[7 Marks]**

Using the sample table, provide a functional dependency which is NOT determined by the primary key.

Note: if one of your dependencies from 3(b) does not use the primary key as the determinant, it is ok to use the same answer again here.

#### **Q 3(d)**

**[9 Marks]**

Using the sample table, provide a functional dependency where  $A$  (left hand side) has 2 or more attributes. Explain what the dependency means and why the degree of  $A > 1$  is necessary.

**QUESTION 4**  
*Normalisation*

**[TOTAL MARKS: 25]**

**Q 4(a)**

**[6 Marks]**

Using your E-R diagram, convert one of your entities (it might be a good idea to use the biggest) to a table which is NOT in 2<sup>nd</sup> Normal form.

Enter 10 rows of data into this table.

Explain why it is not in 2NF and what that means.

**Q 4(b)**

**[7 Marks]**

Show how to fix this problem so that the table is now in 2NF.

**Q 4(c)**

**[6 Marks]**

Using your E-R diagram, convert one of your entities to a table which is in 2NF but NOT in 3<sup>rd</sup> Normal form. [Note: if the result of Q4b is a table in 2NF but not in 3NF, it is ok to use that].

Enter 10 rows of data into this table.

**Q 4(d)**

**[6 Marks]**

Show how to fix this problem so that the table is transformed to 3NF.